

Chapter 3

System Requirements specifications and Analysis

Software Requirements

- A *requirement* is a vital feature of a new system which may include
 - ✓ processing or capturing of data,
 - ✓ controlling the activities of business.
 - ✓ producing information and
 - ✓ supporting the management.

Requirements Engineering

- Requirement engineering is the process to gather the software requirements from client, analyzing and documenting them .
- The goal of requirement engineering is to develop and maintain sophisticated and descriptive 'System Requirements Specification' document.

❑ Requirement Engineering Process is a four step process, which includes

- ❖ Feasibility Study
- ❖ Requirement Gathering
- ❖ Software Requirement Specification
- ❖ Software Requirement Validation

Requirements Engineering

- The software requirements are **description of features** and **functionalities** of the target system.
- Requirements express the expectations of users from the software product.
- The requirements can be **obvious or hidden**, **known or unknown**, **expected or unexpected** from client's point of view..

Requirements Engineering

❑ Feasibility Study

- the analysts does a detailed study about **whether the desired system and its functionality are feasible to develop**.
- This study analyzes **whether the software product can be practically materialized in terms of implementation, contribution of project to organization, cost constraints and as per values and objectives** of the organization

❑ Requirement Gathering

- If the feasibility report is **positive** towards undertaking the project, next phase starts with **gathering requirements** from the user.
- Analysts and engineers **communicate** with the client and end-users to know their ideas on **what the software should provide** and **which features they want the software** to include.

Requirements Engineering

❑ Software Requirement Specification

- **SRS** is a document created by system analyst after the requirements are collected from various stakeholders.
- It defines **how** the intended software **will interact** with **hardware**, **external interfaces**, **speed of operation**, **response time of system**, portability of software across various platforms, **maintainability**, **speed of recovery after crashing**, **Security**, **Quality**, **Limitations** etc.

Requirements Engineering

➤ SRS should come up with following features:

- ✓ User Requirements are expressed in natural language.
- ✓ Technical requirements are expressed in structured language, which is used inside the organization.
- ✓ Design description should be written in Pseudo code.
- ✓ Format of Forms and GUI screen prints.
- ✓ Conditional and mathematical notations for DFDs etc.

Requirement Engineering

❑ Software Requirement Validation

- After requirement specifications are developed, the requirements mentioned in the document should be validated.
 - User might ask for **illegal**, impractical solution or **experts may interpret the requirements incorrectly**.

Requirements can be checked against following conditions :

- ❖ If they can be practically implemented
- ❖ If they are valid and as per functionality and domain of software
- ❖ If there are any ambiguities
- ❖ If they are complete
- ❖ If they can be demonstrated

Types of Requirements

❑ **Functional Requirements**

- These are the requirements that **the end user specifically demands** as basic facilities that the **system should offer**.
- All these functionalities need to be necessarily **incorporated** into the **system** as a part of the contract.
- These are represented or **stated in the form of input** to be given to the system, the **operation** performed and the **output** expected.

Types of requirements

❑ **Non-functional requirements**

- These are basically the quality constraints that the system must satisfy according to the project contract.
- The priority or extent to which these factors are implemented varies from one project to other.
- They basically deal with issues like:
 - ***Portability, Security, Maintainability, Reliability, Scalability, Performance, Reusability, Flexibility....***

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- Non-functional Requirements are classified into following types:
- Interface constraints
- **Performance constraints**: response time, security, storage space, etc.
- **Operating constraints**
- **Life cycle constraints**: maintainability, portability, etc.
- **Economic constraints**
- The process of specifying non-functional requirements requires the **knowledge of the functionality** of the system, as well as the **knowledge of the context** within which the system will operate.

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❑ Domain requirements

- Domain requirements are the requirements which are characteristic of a particular category or domain of projects.
- The basic functions that a system of a specific domain must necessarily display come under this category.
- *For instance, in an academic software that maintains records of a school or college, the functionality of being able to access the list of faculty and list of students of each grade is a domain requirement.*
- These requirements are therefore identified from that domain model and are not user specific

Activities in requirement Determination

➤ **Requirements Anticipation**

- It predicts the characteristics of system based on previous experience which include certain problems or features and requirements for a new system.

➤ **Requirements Investigation**

- It is studying the current system and documenting its features for further analysis.

➤ **Requirements Specifications**

- It includes the analysis of data which determine the requirement specification, description of features for new system, and specifying what information requirements will be provided.

Requirement finding techniques

- The main aim of fact finding techniques is to determine the information requirements of an organization used by analysts to prepare a precise SRS understood by user.

□ Interviewing

- Systems analyst collects information from individuals or groups by interviewing.
- The analyst can be formal, legalistic, play politics, or be informal; as the success of an interview depends on the skill of analyst as interviewer.
- **Unstructured Interview** – The system analyst conducts question-answer session to acquire basic information of the system.
- **Structured Interview** – It has standard questions which user need to respond in either close (objective) or open (descriptive) format.

Requirement finding techniques

❖ **Advantages of Interviewing**

- This method is frequently the best source of gathering qualitative information.
- It is useful for them, who do not communicate effectively in writing or who may not have time to complete questionnaire.
- Information can easily be validated and cross checked immediately.
- It can handle the complex subjects.
- It is easy to discover key problem by seeking opinions.
- It bridges the gaps in the areas of misunderstandings and minimizes future problems.

Requirement finding techniques

❑ Questionnaires

- This method is used by analyst to gather information about various issues of system from large number of persons.
- There are two types of questionnaires –
 - **Open-ended Questionnaires** – It consists of questions that can be easily and correctly interpreted. They can explore a problem and lead to a specific direction of answer.
 - **Closed-ended Questionnaires** – It consists of questions that are used when the systems analyst effectively lists all possible responses, which are mutually exclusive.

Requirement finding techniques

❖ **Advantages of questionnaires**

- It is very effective in surveying interests, attitudes, feelings, and beliefs of users which are not co-located.
- It is useful in situation to know what proportion of a given group approves or disapproves of a particular feature of the proposed system.
- It is useful to determine the overall opinion before giving any specific direction to the system project.
- It is more reliable and provides high confidentiality of honest responses.
- It is appropriate for electing factual information and for statistical data collection which can be emailed and sent by post.

Requirement finding techniques

❑ Review of Records, Procedures, and Forms

- Review of existing records, procedures, and forms helps to seek insight into a system which describes the current system capabilities, its operations, or activities.

❖ Advantages

- It helps user to gain some knowledge about the organization or operations by themselves before they impose upon others.
- It helps in documenting current operations within short span of time as the procedure manuals and forms describe the format and functions of present system.
- It can provide a clear understanding about the transactions that are handled in the organization, identifying input for processing, and evaluating performance.
- It can help an analyst to understand the system in terms of the operations that must be supported.
- It describes the problem, its affected parts, and the proposed solution.

Requirement finding techniques

❑ **Observation**

- This is a method of gathering information by noticing and observing the people, events, and objects. The analyst visits the organization to observe the working of current system and understands the requirements of the system.

❖ **Advantages**

- It is a direct method for gleaned information.
- It is useful in situation where authenticity of data collected is in question or when complexity of certain aspects of system prevents clear explanation by end-users.
- It produces more accurate and reliable data.
- It produces all the aspect of documentation that are incomplete and outdated.

Requirement finding techniques

❑ Joint Application Development (JAD)

- It is a new technique developed by IBM which brings owners, users, analysts, designers, and builders to define and design the system using organized and intensive workshops. JAD trained analyst act as facilitator for workshop who has some specialized skills.

❖ Advantages of JAD

- It saves time and cost by replacing months of traditional interviews and follow-up meetings.
- It is useful in organizational culture which supports joint problem solving.
- Fosters formal relationships among multiple levels of employees.
- It can lead to development of design creatively.
- It Allows rapid development and improves ownership of information system.

Requirement finding techniques

❑ Secondary Research or Background Reading

- This method is widely used for information gathering by accessing the gleaned information. It includes any previously gathered information used by the marketer from any internal or external source.

❖ Advantages

- It is more openly accessed with the availability of internet.
- It provides valuable information with low cost and time.
- It act as forerunner to primary research and aligns the focus of primary research.
- It is used by the researcher to conclude if the research is worth it as it is available with procedures used and issues in collecting them.

Requirement Analysis

□ Requirement Analysis

- ways to accurately **capture, interpret, and represent the voice of customers** when specifying the requirements for a software product.
- Requirements analysis **is critical to the success or failure of a systems** or software project.
- The requirements should be **documented, actionable, measurable, testable, traceable**, related to identified business needs or opportunities, and defined to a level of detail sufficient for system design

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☐ **Activities of requirements analysis includes**

1. Eliciting requirements:

☐ the task of communicating with customers and users to determine what their requirements are.

☐ Requirement elicitation illustrates the following



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- **Requirements gathering** - The developers discuss with the client and end users and know their expectations from the software.
- **Organizing Requirements** - The developers prioritize and arrange the requirements in order of importance, urgency and convenience.
- **Negotiation & discussion** - If requirements are ambiguous or there are some conflicts in requirements of various stakeholders, it is then negotiated and discussed to remove the ambiguity and conflicts, they are discussed for clarity and correctness.
- **Documentation** - All formal & informal, functional and non-functional requirements are documented and made available for next phase processing.

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❑ *Activities of requirements analysis includes*

2. Analyzing requirements: determining whether the stated requirements are unclear, incomplete, ambiguous, or contradictory, and then resolving these issues.

3. Requirements modeling: Requirements might be documented in various forms, such as natural-language documents, use cases, user stories, or process specifications.

4. Review and retrospective: Team members reflect on what

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- Once the requirements are gathered, we document the requirements in a **Software Requirements Specification (SRS)** document, use cases or as User Stories, which are shared with the stakeholders for approval
- This document is **easy to understand** for both **normal users** and **developers**.
- Any changes in the requirements are also documented and go through a change control procedure and finalized on approval

Structured analysis vs Object oriented analysis

- Analysts use various tools to understand and describe the information system.

❑ **Structured Analysis**

- It is a development method that allows the analyst to understand the system and its activities in a logical way.
- It is a systematic approach, which uses graphical tools that analyze and refine the objectives of an existing system and develop a new system specification which can be easily understandable by user.

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➤ It has following attributes –

- It is **graphic which specifies the presentation** of application.
- It **divides the processes** so that it gives a clear picture of system flow.
- It is **logical rather than physical** i.e., the elements of system do not depend on vendor or hardware.
- It is an approach that works from **high-level overviews to lower-level details.**

Structured analysis vs Object oriented analysis

❑ **Structured Analysis Tools**

- During Structured Analysis, various tools and techniques are used for system development. They are –
 - Data Flow Diagrams
 - Data Dictionary
 - Decision Trees
 - Decision Tables
 - Structured English
 - Pseudocode

Structured analysis vs Object oriented analysis

❑ **Object oriented analysis**

- In the object-oriented approach, the focus is on capturing the structure and behavior of information systems into small modules that combines both data and process.
- The main aim of Object Oriented Design (OOD) is to improve the quality and productivity of system analysis and design by making it more usable.

Structured analysis vs Object oriented analysis

❑ **Object oriented analysis**

- The OO model is beneficial in the following ways –
 - It facilitates changes in the system at low cost.
 - It promotes the reuse of components.
 - It simplifies the problem of integrating components to configure large system.
 - It simplifies the design of distributed systems.